

Adaptive Lossless Compression of Floating-Point Time-Series Data Using Enhanced XOR-Based Encoding

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ABSTRACT:

In today's data-driven world, the massive growth of time-series data generated by IoT devices, industrial sensors, and monitoring systems poses significant challenges in terms of storage, transmission, and processing efficiency. Managing such large-scale datasets efficiently is essential for real-time analytics, predictive maintenance, and intelligent decision-making. This project presents a novel approach for efficient lossless compression of floating-point time-series data through an enhanced algorithm named ACTF (Adaptive Coding and Temporal Feature-based compression). Building upon existing methods such as Gorilla and Chimp, the proposed algorithm introduces three major enhancements-data preprocessing, extended coding categories, and refined feature representation-to achieve superior compression performance. In our system, incoming time-series data is first analyzed to detect redundancy and equality among neighboring values. The ACTF algorithm then applies adaptive encoding patterns that reduce branch lengths and improve the handling of leading and trailing zeros. These refinements minimize redundancy while maintaining data fidelity. Experimental evaluation on diverse IoT and server datasets demonstrates that ACTF achieves higher compression ratios and faster compression and decompression times compared to state-of-the-art algorithms such as Gorilla, Chimp, and Chimp128. This approach provides a cost-effective and scalable solution for managing large volumes of time-series data while ensuring high efficiency and data integrity. The combination of adaptive preprocessing and advanced encoding techniques makes ACTF an ideal choice for modern data-intensive applications, offering improved storage optimization, reduced transmission overhead, and enhanced performance in real-time systems.